CS3000 Algorithms & Data **Practice Quiz 5C**	**Name:** _____ **NUID:** _____

Instructions:

- Please put your phones on silent and in your bag for the duration of the quiz.
- This quiz is designed to last 40 minutes.
- If you have a question, alert a proctor quietly, and we'll be glad to assist.
- If we can't read your answer, we can't grade your answer. Please write legibly.
- Practice version: slightly harder than the real quiz; both are new problems in the style of the notes' practice list.

Q1. Every positive integer can be written as a sum of perfect squares, e.g. $12 = 4 + 4 + 4$. Let $S(n)$ be the fewest perfect squares that sum to n.

(a) Write a recurrence for $S(n)$ with its base case. (Hint: what is the last square used?)

(b) Fill in the table bottom-up for $n = 0, \ldots, 13$.

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13
$S(n)$														

(c) Greedy (take the largest square that fits, repeat) works for $n = 13$. Find the smallest n in your table where greedy is *not* optimal, and show both answers.

(d) Write bottom-up pseudocode for $S(n)$ and give its running time in terms of n.

Q1 (continued).

Q2. Your professor posts optional assignments. Assignment A_i takes h_i hours and earns p_i extra-credit points, and each can be done at most once. You need **at least** $P = 8$ points and want to work as few hours as possible:

Assignment	A_1	A_2	A_3	A_4
Hours h_i	5	2	4	3
Points p_i	6	3	5	4

Let $H(i, k)$ be the minimum hours needed to earn at least k points using only $A_1, \ldots, A_i$ ($H = \infty$ if impossible).

(a) Write the recurrence for $H(i, k)$ with base cases. (Careful: what happens when $p_i \geq k$?)

(b) Fill in the table bottom-up.

$i \downarrow \; k \rightarrow$	0	1	2	3	4	5	6	7	8
$i = 0$									
$i = 1$									
$i = 2$									
$i = 3$									
$i = 4$									

(c) What is the minimum number of hours, and which assignments achieve it? Show the traceback.

(d) How is this problem like 0/1 knapsack, and how is it different?

Q2 (continued).